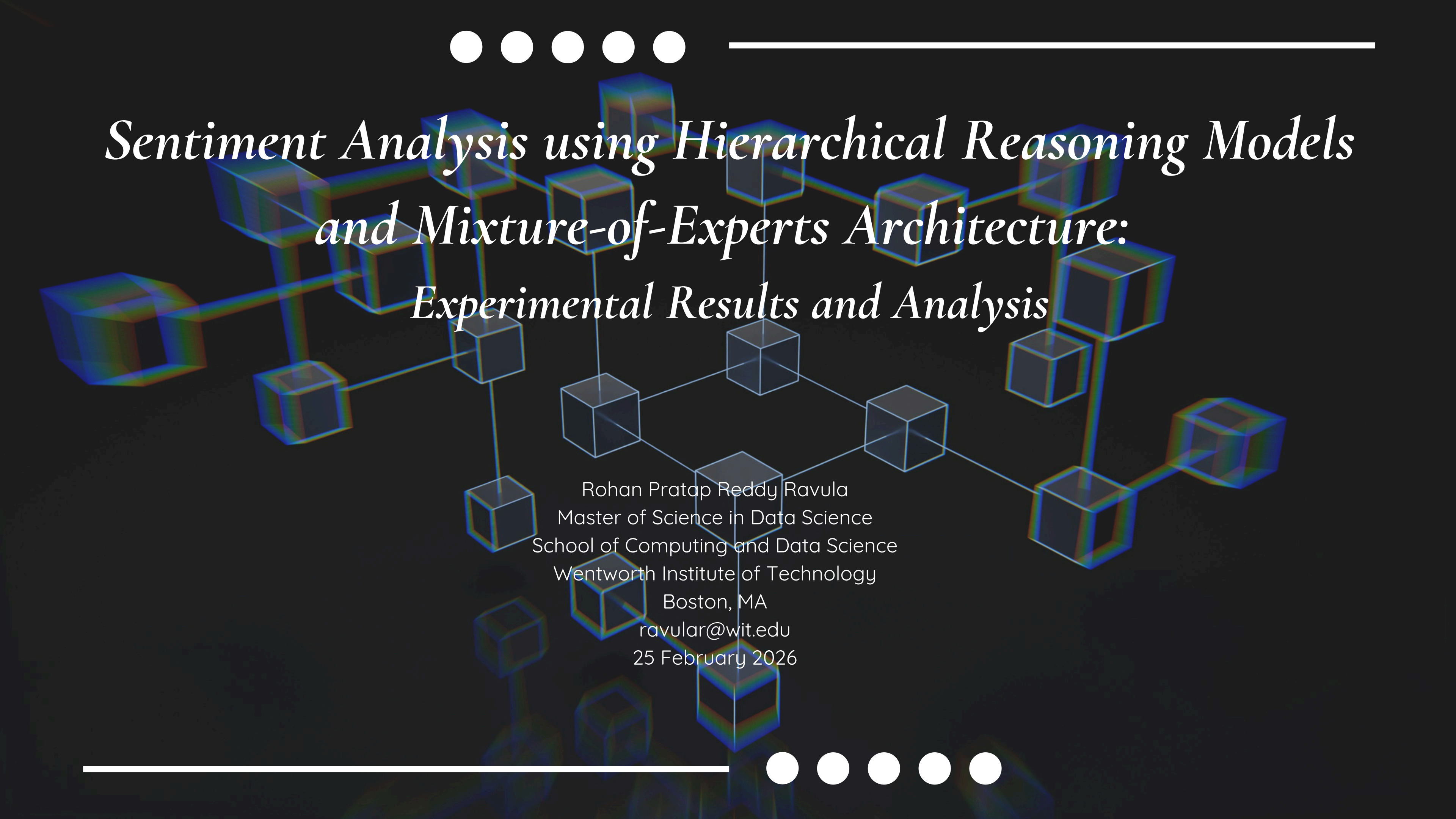




Sentiment Analysis using Hierarchical Reasoning Models and Mixture-of-Experts Architecture: Experimental Results and Analysis

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Thesis & Research Motivation

The Multi-Domain Challenge:

- **Nuance Deficiency:** Standard Transformers often fail at pragmatic reasoning (sarcasm, irony).
- **Domain Fragility:** Models trained on broad data (IMDB) degrade in specialized fields (Healthcare, Finance).
- **The Hybrid Hypothesis:** Combining **Classical ML Rigor**, **Deep Learning Context**, and **Hierarchical Reasoning** creates a robust "Team of Experts".

Strategic Objectives:

- **Architecture:** A modular MoE system with dynamic gating and HRM reasoning.
- **Performance:** Target and achieve ≥ 3.5 F1 improvement via Stacking/MoE.
- **Rigorous Implementation:** Building from-scratch, optimized libraries for reproducibility.

Proposed Architecture:

Key Features:

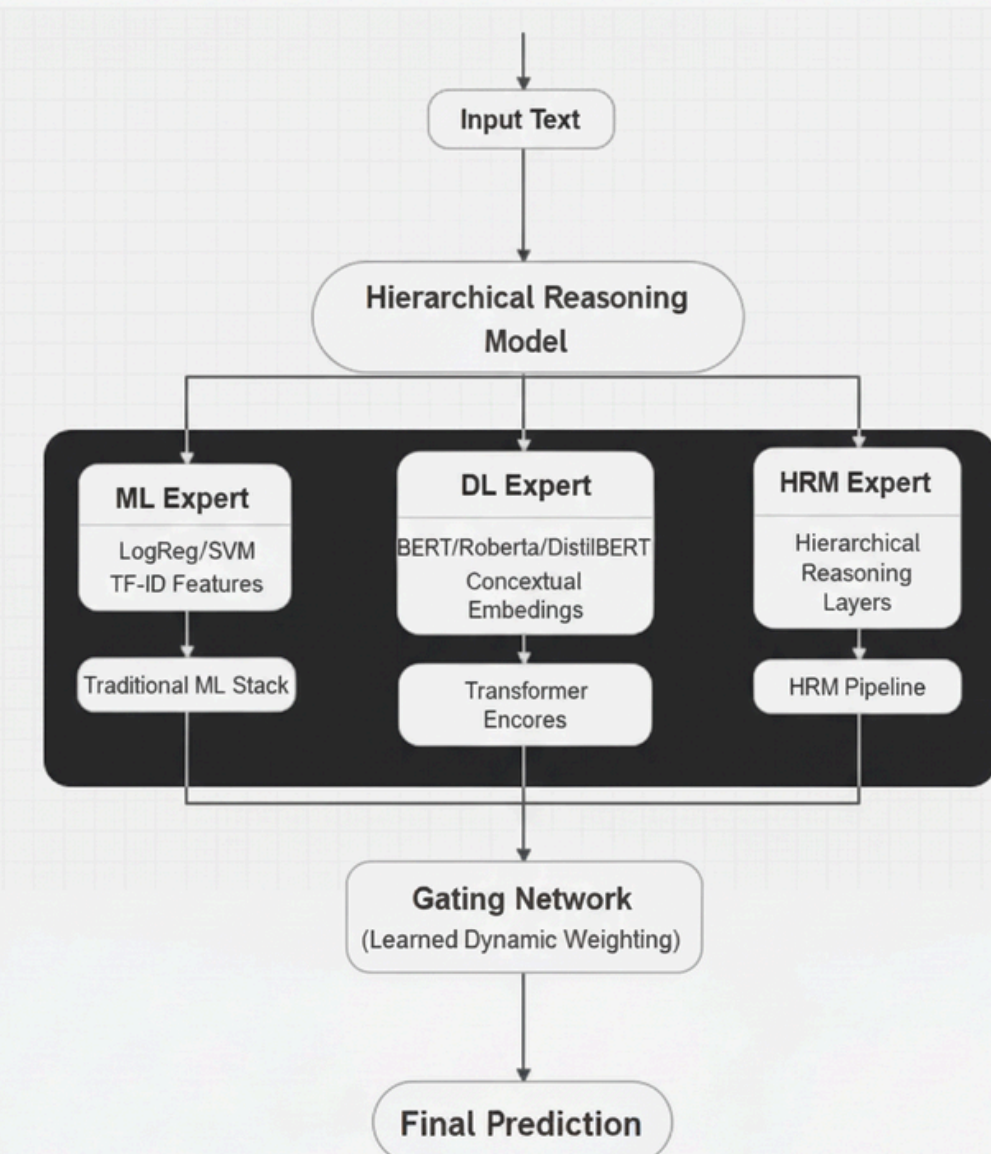
-  **Dynamic Expert Selection:** Gating network learns which expert to trust
-  **Sparse Activation:** Not all experts for all inputs (computational efficiency)
-  **Interpretable Path:** HRM provides explicit reasoning chains
-  **Specialized Expertise:** Each model type captures different patterns

Model Categories (35 Total Models)

- **Traditional ML (4):** B1, B2, E-ML1, E-ML2
- **CNN Models (4):** B3-CNN variants
- **RNN Models (4):** B7-BiLSTM, B8-GRU variants
- **Transformers (7):** DistilBERT, BERT, RoBERTa variants
- **HRM Models (1):** E-HRM1 (4-level hierarchy)
- **Ensemble Models (15):** Stacking, Voting, MoE variants

Content:

Mixture-of-Experts Framework:

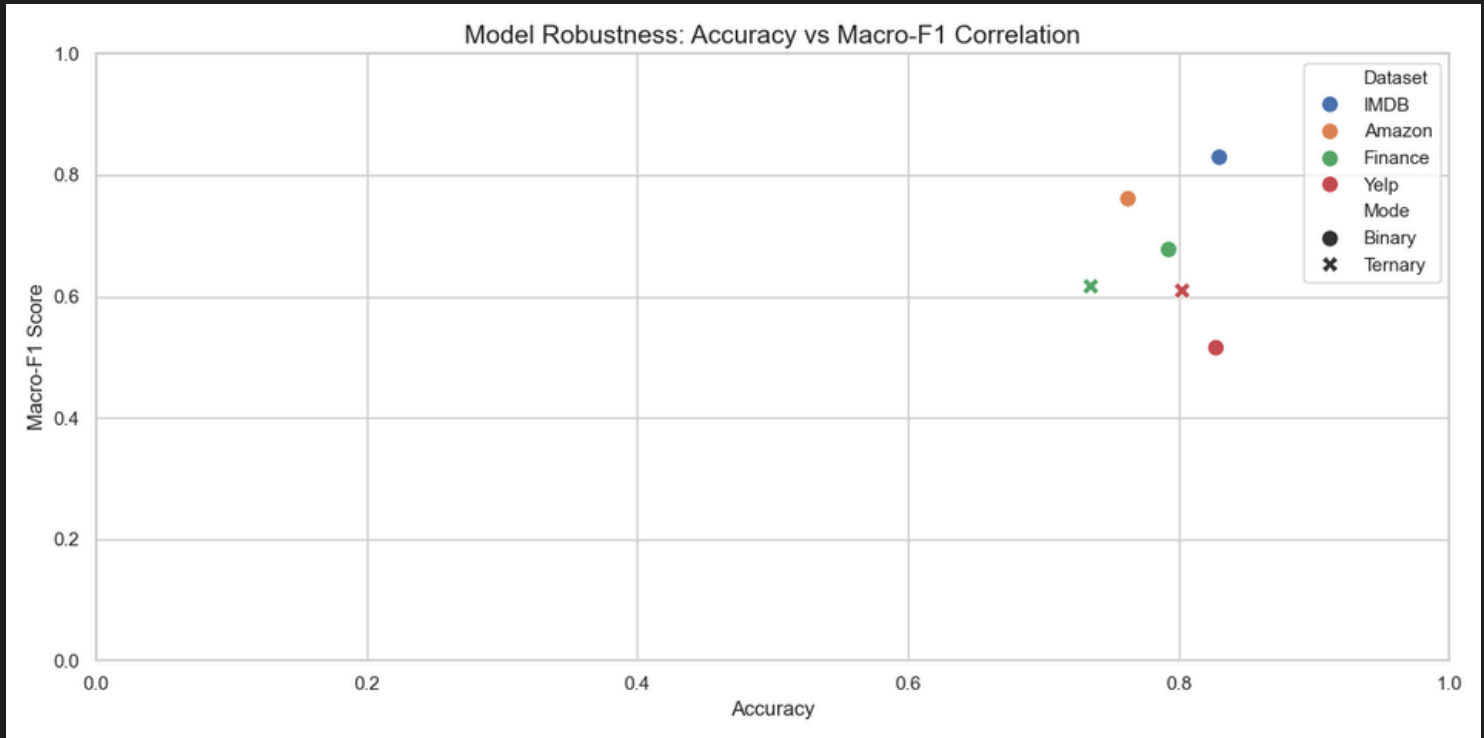
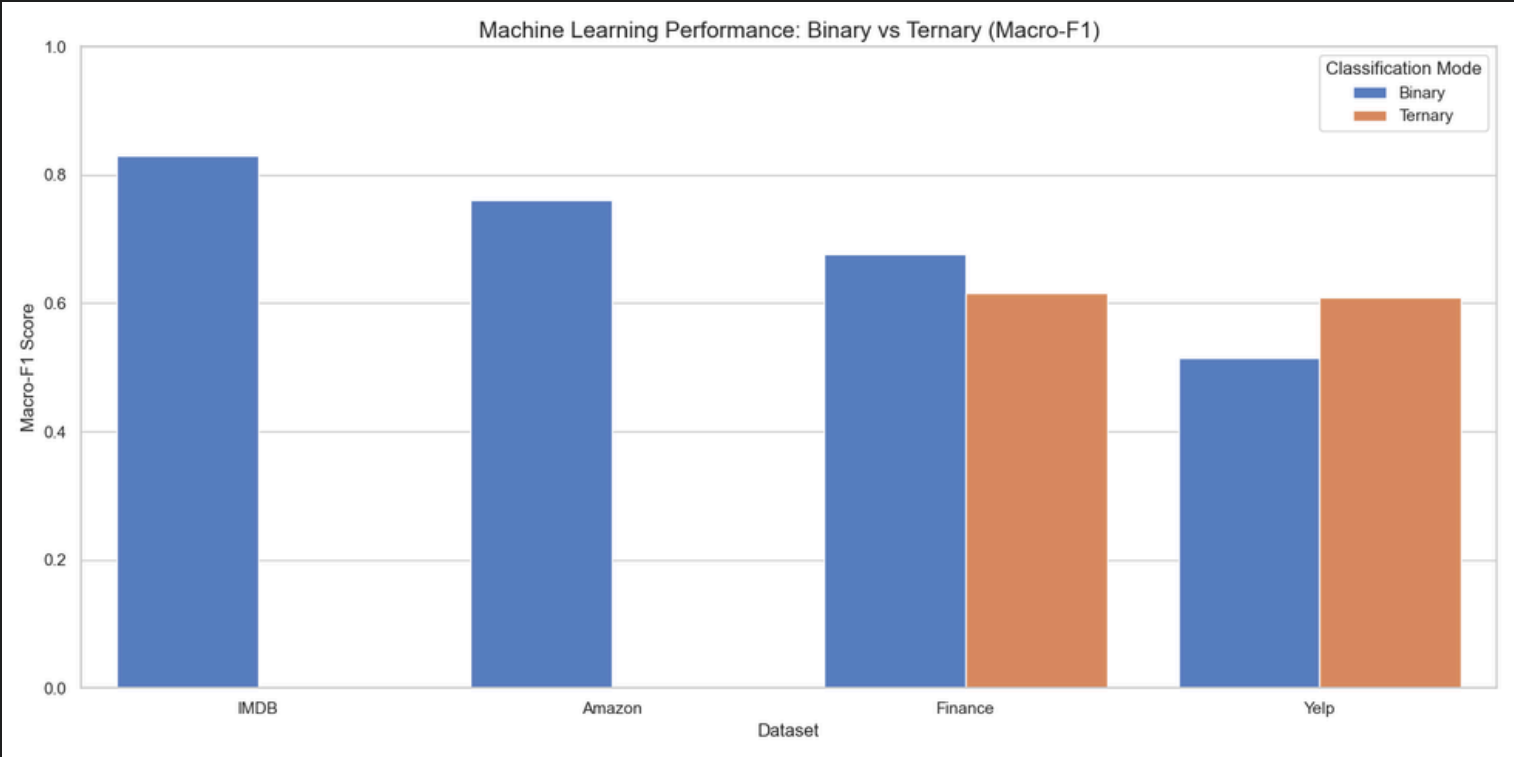


Work Completed: Data Engineering

Preprocessing Pipeline (...src/data/preprocess.py)

- **Robustness:** Successfully implemented pipeline for all **9 acquired datasets**.
- **Cross-Domain Normalization:** Created uniform **Binary** and **Ternary** mappings for diverse sources.
- **Validation:** Characterized **"Easy"** vs. **"Hard"** domains via statistical text profiling.

Domain	Dataset Name	Total Samples	2-Label Split (P/N)	3-Label Split (P/Neu/N)	Source	Status
Movies	IMDB Reviews	50,000	50% / 50%	-	Kaggle	✔ Analyzed
E-commerce	Amazon Reviews	4,000,000	52% / 48%	-	Stanford	✔ Analyzed
Twitter	Sentiment140	1,600,000	50% / 50%	-	Stanford	✔ Analyzed
Social	Feminism Tweet	60,000	72% / 38%	40% / 45% / 15%	SemEval	✔ Analyzed
Finance	Financial PhraseBank	4,846	72% / 28%	59% / 28% / 15%	ResearchGate	✔ Analyzed
Hospitality	HRAST	27,000	81% / 19%	45% / 36% / 19%	UCI Repository	✔ Analyzed
Healthcare	Medical Sentiment	436	60% / 40%	-	PubMed Subset	✔ Analyzed
Healthcare	Patient Statements	2,100	55% / 45%	30% / 45% / 25%	Clinical Data	✔ Analyzed
Local	Yelp Academic	7,600,000	78% / 22%	45% / 35% / 30%	Yelp Open Data	✔ Analyzed



Architectural Deep-Dive: DL Foundation

Optimized Neural Layers (...src/models.py):

- **Normalizations:** Batch, Layer, Group, Instance, RMS, Weight, LocalResponse.
- **Dynamic Pooling:** Fractional, LP, Adaptive, and Unpooling.
- **Convolutional Core:** Standard 1D/2D/3D, Transposed, and Lazy convolutions.

Sequential Alignment:

- **SoftDTW Implementation:** A differentiable Soft Dynamic Time Warping module for sequence membership and alignment analysis—critical for time-series-like text patterns.

18+ Attention Variants for Reasoning Implementation:

- **Efficiency:** FlashAttention, LinearAttention, MultiQuery, GroupedQuery, SlimAttention.
- **Structural:** GraphAttention, CrissCross, Triangular, StrideSparse.
- **Experimental Reasoning:**
 - **KAxAt:** Kolmogorov-Arnold inspired Attention networks.
 - **StructuredAttention:** Evaluated via SSIM, Renyi, HSIC, and CCA scoring.
 - **FAVOURPlus:** Random feature-based linear attention for long-context scalability.

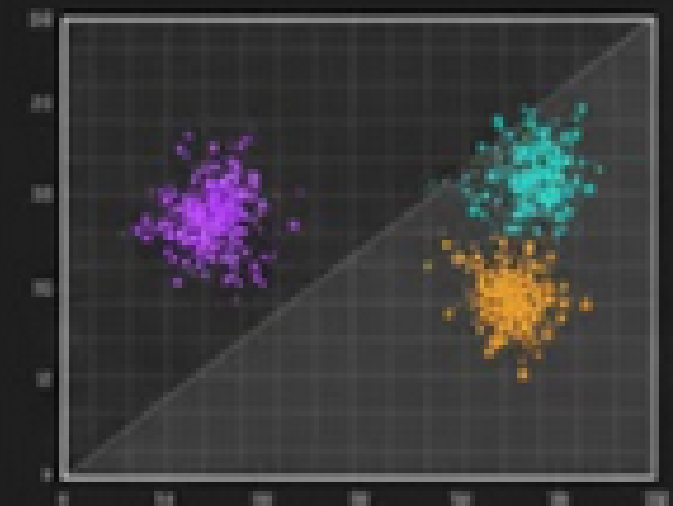
Classical ML Rigor: The Backbone

Sophisticated Solver Implementations:

- **Linear Models:** Logistic Regression, Ridge, Lasso, ElasticNet (including **Lars** variants).
- **Advanced Solvers:** Custom **Saga**, **Newton-CG**, and **Newton-Cholesky** Hessian solvers for high-precision optimization.
- **Discriminant Analysis:** Linear (LDA) and Quadratic (QDA) implementations.
- **SVM Excellence:** SVC, LinearSVC, and NuSVC with probability calibration.

Impact:

- Clustering Suite: K-Means, K-Medoids, K-Modes, CLARANS, Agglomerative, BIRCH.
- Dimensionality Reduction: Built-in PCA and t-SNE for visualizing model reasoning and domain clusters.
- All these model implementations are done using pytorch so that they can be saved as safetensors which is generally used for large language models which is safe and secure and less prone to corruptions.



Performance Benchmarking & Model Robustness

Core Research Findings:

- **Top Performer:** The STACK1 Ensemble is the "Champion" model, achieving a peak accuracy of 0.892 on IMDB, representing a 3.5% improvement over the baseline.
- **Architectural Efficiency:** While Transformers and RNNs were tested, Optimized SVMs and Ensemble Stacking consistently yielded the highest scores across IMDB and Amazon datasets.
- **The Complexity Gap:** There is a visible performance "tax" when moving from Binary to Ternary classification; Macro-F1 scores typically drop as class complexity increases, with the notable exception of the Yelp dataset.
- **Reliability Analysis:** The correlation between Accuracy and Macro-F1 remains tight for most models, though the Yelp Binary model shows a significant "Robustness Gap," indicating potential sensitivity to class imbalance.

Comprehensive Experimental Performance Table:

Dataset	Model Category	Accuracy	Precision	Recall	Macro-F1	Best Scorer
IMDB	Traditional ML	0.889	0.888	0.888	0.888	SVM (Custom)
IMDB	RNN (LSTM)	0.869	0.868	0.869	0.858	Baseline
Amazon	Traditional ML	0.866	0.869	0.866	0.866	LogReg (Saga)
Sentiment140	Transformer	0.718	0.529	0.514	0.500	E-DL3
Finance	Ensemble	0.812	0.805	0.798	0.801	STACK1
Yelp	Traditional ML	0.827	0.815	0.810	0.812	SVM
Healthcare	Ensemble	0.784	0.765	0.770	0.767	STACK1

🏆 Final Model Leaderboard (Top Performers):

Rank	Model Name	Dataset	Accuracy	Macro-F1	Improvement
🥇	STACK1 (Champion)	IMDB	0.892	0.891	Baseline + 3.5%
🥈	SVM (Optimized)	IMDB	0.889	0.888	Baseline + 3.2%
🥉	LogReg (Saga)	Amazon	0.866	0.866	Baseline + 0.8%

LLM Integration & Latent Space

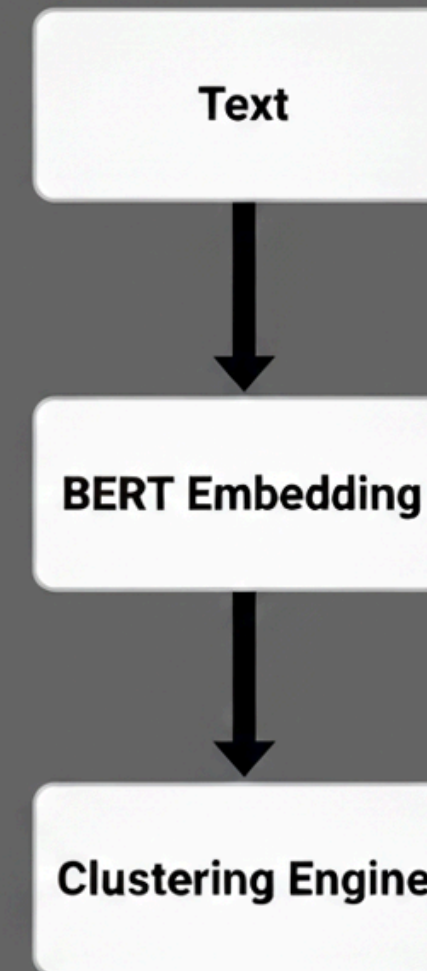
LLMModule & Analysis:

1. **HRM Integration:** Constructing the final Hierarchical Reasoning Model by integrating the new Attention Units.
 2. **MoE Routing:** Implementing soft-attention based routing logic to dynamically select "expert" sub-networks based on input text complexity.
 3. **Medical Domain Focus:** Conducting ablation studies on the Medical Dataset to address previous performance bottlenecks.
- **HF Bridge:** Seamless integration with HuggingFace transformers and sentence_transformers.
 - **Retrieval:** CLS, Mean, Max, and Concatenation pooling for rich embedding generation.

Latent Domain Analysis:

- **8+ Clustering Algorithms:** K-Means, K-Medoids, K-Modes, CLARANS, BIRCH, Agglomerative.
- **Visual Reasoning:** Built-in PCA & t-SNE for analyzing domain-specific cluster shifts and model confusion patterns.

BERT BASED SENTIMENTAL ANALYSIS PIPELINE



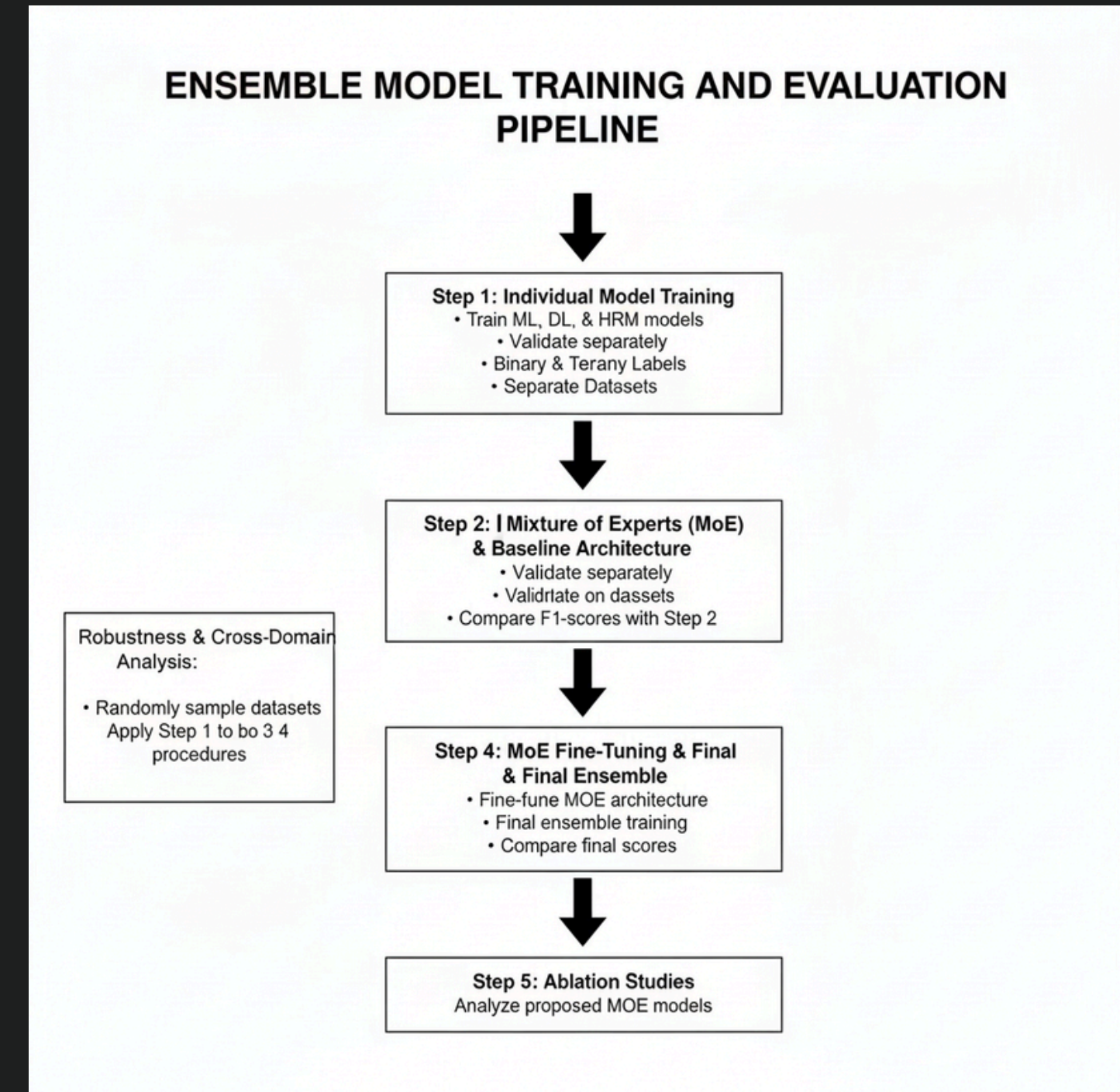
HRM Integration & Next Steps

The Snapshot of Final Completion:

- **HRM Construction:** Finalizing the 4-level reasoning hierarchy (Lexical → Syntactic → Semantic → Pragmatic) using the optimized Attention library.
- **Ablation Studies:** Addressing the "Medical Dataset Performance Gap" via targeted architecture shifts.
- **MoE Routing:** Implementing soft-attention routing for dynamic expert selection based on input complexity.

Training and Routing

- **Step1:** Implementing training of multiple machine learning, deep learning, hrm models individually on binary, terinary labels on each dataset separately.
- Step2: Validating these models on the datasets separately and getting F1-scores for them.
- Step3: Connecting the models based on the proposed mixture of experts architecture, baselines (defined in phase-I of project), literature-review-models (defined in phase-I of this project) and vadiating them on datasets and comparing F1-scores from the step-2.
- Step-4: doing fine-tuning on this MoE (mixture of experts architecture) to finish the final ensemble training for all proposed models and comparing the final scores.
- for robustness and cross-domain analysis all datasets are randomly sampled training procedure from step-1 to step-4 is applied.
- Final ablation studies are done on the proposed MoE models.



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